The Questions

Answers: The answers are indicative , they do not have to be exactly touching these points , may include some others aspects

1.

a)

i) List one dynamic event with its defining characteristic that can be classed as wave phenomenon in power systems.

[2]

Ans: Lightning or switching . If lightning , the characteristcis is tens of kA with microsecond rise time and tens of microseconds as fall time. If switching , it is in hundrededs of kV, and tens of milliseconds.

ii) Briefly explain the influence of this dynamic event on the stability/control/protection design considerations in power systems.

[4]

Ans: The lightning strike in transmission circuit/transformer leads to large temporary currents on the components. They can propagate further to create fault. The impulse current gives rise to impulse voltage . So the control/protection in the form of pilot wire over the power lines , lightning arrestors for protecting transformers, reactor, capacitors are used. The insulation design must consider basic impulse level voltage . For switching related phenomena surge over voltage protection for switch gear circuit breaker , transformer, reactors are needed. Rate of rise of restriking voltage (RRRV) is also a design consideration. Here insulation design and coordination also becomes very important.

b) Normally stator transients are neglected in transient or first rotor swing stability in the context of large synchronous generation:

i) Why is it so?

[2]

Ans: The rotor swings slowly as it is large mass , it is electromechanical in nature . The transients in the stator circuits are electromagnetic in nature , usually 50 Hz and above. While we focus on first swing stability , the time scale of stator transients are very small (ms) comapared to time scale of observation of rotor swings . The overall impact on the stator circuit voltage phasor (RMS) from the stator transients is small but always positive . So only the speed voltage dynamics are considered which is driven by rotor flux dynamics, neglecting fast stator transients or transformer voltage in the stator circuits.

ii) How will the first swing stability qualitatively/quantitively be impacted when these are included ?

[3]

Ans: The inclusion of stator transient means the transformer voltage terms will be included on the behaviour of d-q axis phasor quantities. The rotor angle will have power frequency (50/60 Hz) modulation. The electromagnetic torque will be modified with some ripple . Because stator has resistance , there will be damping effect . Angle swing will not be as

high as without inclusion of this component in the model. So qualitatively it will be more optimistic and quantiatively it will be swing with less excursion.

iii) How are these stator transients modelled for electromechanical stability studies involving torsional stability in multi-stage turbine mass ?

[2]

Ans: They are modelled as differential equations with stator inductance and resistance included because the torsional stability involving multi-mass turbine generator has natural frequency spanning from 10-45 Hz. The time scale of stator transients are comaparable with the time scale of torsional dynamics.

iv) Why can not these transients be ignored for the stability analysis of Type 3 /Type 4 wind turbines?

[3]

Ans: Type 3 and Type 4 wind turbines have power converter and control. The mass associated with Type 3 rotor is not as huge as that of a large centralised synchronous genertors. The control bandwidth of the converters are of relatively of higher frequency . The electromagnetic transients of stator windings are very close in time scale with those . So they can not be ignored.

c)

i) How do the laminated stator and solid rotor of a synchronous machine help power system stability ?

[4]

Ans : Laminated stator does not impact much electromechanical stability, but because it minimises heating due to magnetic losses (hysteresis and eddy current) , it provide thermal stability to stator structure . The solid rotor offers good path for circulating currents, during rotor swing. It offers damping against oscillatory instability, particularly at plant mode/local mode.

ii) Why do the reactances of stator windings drop suddenly when they start feeding fault currents ?

[3]

Ans: The rotor field opposes any changes in the flux in its path when additional magnetic field appears due to large stator currents because of fault. The flux largely flows through high reluctance air . This results in lower inductance . After sometime , the flux created by the stator MMF is allowed to flow through peripherial surface of the rotor increasing the reactance further before reaching steady state value which is high.

iii) Why is it not advisable to operate a synchronous generator in leading power factor (VAR absorption) mode from a stability perspective?

[2]

Ans: In leading power factor mode stator and rotor flux have additive alignment. The end part (overhang) of the stator winding are bound for structural rigidity through an end ring to tighten and fix its position against any vibration. The end ring is conductive , so power frequency voltage is induced resulting in circulating current within the body of the end rings. This leads to the expansion of rings due to heating. The overhang looses strength due to binding force. Also the synchronising power co-efficent during leading power factor operation is reduced. For the same amount of MW output , load angle during leading power factor operation is more than that during lagging power factor operation. So angle stability margin in steady state is less in leading power factor operation.

[25 marks]

2.

a)

i) What is non-minimal phase transfer function in the contexts of hydro turbines?

[3]

Ans: A non minimal phase transfer function has a right half plane zero. In the hydro turbine dynamics's context, it is the instantenous reverse proportional relationship between input (gate opening) to the output (mechanical power produced by the turbine).

ii) How does it impact the stability of speed governing for frequency response ?

[3]

Ans: Right half plane zero reduces phase as well as gain margin at high frequency. For any fast change , the output drops which is not desirable as the purpose of gate opening is to prodoce more power to provide stable frequency. So, very Fast speed governance is not possible .

iii) What control strategies are adopted to counter any adverse impact of the non-minimal phase transfer function behaviour?

[2]

Ans: Usually transient gain reduction (TGR) mechanisms are implemented with suitable phase lag circuits.

b) A round rotor star connected synchronous generator is connected to the grid at 15.75 kV (line to line) voltage. The rated power is 210 MW, power factor is 0.85 (lagging). The synchronous reactance is 1.7 pu on its rated base of 247 MVA. The resistance is negligible. Find out:

i) the rated phase current in kA

[2]

ii) the excitation phase voltage in kV

[3]

iii) load angle (δ) in degrees

[2]

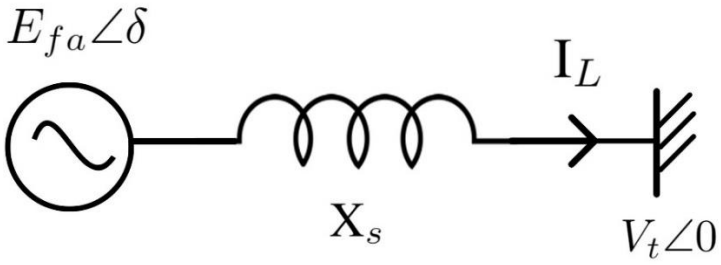
iv) load angle stability margin in degrees

[1]

v) total terminal reactive power in MVAR

[3]

Ans :



$V_t = 15.75\ kV, \quad P_{rated} = 210\ MW, \quad Rated\ MVA = 247$
$Power\ factor = 0.85\ lag$
$I_L = \frac{MVA}{\sqrt{3}V_t} = 9054\ A$

$E_{fa} = V_t + j\sqrt{3}I_L X_s$
$Base\ Impedance = \frac{kV^2}{MVA}, \quad Power\ factor\ angle = -cos^{-1}0.85 = -31.78°$
$X_s\ in\ \Omega = X_s\ (p.u.)\frac{kV^2}{MVA} = 1.7 \times 1.0043 = 1.7073\ \Omega$
$E_{fa} = 37.54\ kV\ \angle 37.32°$
$E_{fa}/phase = 21.67\ kV$
$load\ angle\ delta = 37.32°$
$Stability\ Margin = 90° - 37.32° = 52.68°$
$Reactive\ Power = \frac{E_{fa}}{X_s}V_t\ cos\ \delta - \frac{{V_t}^2}{X_s} = 130MVAr$

c) The generator from part (b) is still considered. The excitation voltage is reduced by 10% while mechanical input remains unchanged.

i) Find out the change in steady state stability margin in degrees with respect to the rated excitation voltage.

[3]

Real power does not change. So:

$$P = \frac{0.9E_{fa}V_t}{X_s} sin\ \delta'$$
$$\delta' = 42.34°$$
$$Stability\ Margin = 47.65°$$

ii) Find out the change in total terminal reactive power with respect to the operation at rated excitation voltage.

[3]

$$Reactive\ Power\ = \frac{0.9E_{fa}}{X_s}V_t\ cos\ \delta' - \frac{{V_t}^2}{X_s} = 85MVAr$$

Change in reactive power output is : 130-85 = 45 MVAr

[25 marks]

Real power does not change. So:

$$P = \frac{0.8E_{fa}V_t}{X_s} \sin\delta'$$

$$\delta' = 46.5°$$

$$Stability\ Margin\ =\ 43.5°$$

3.

a)

i) Why are variables (current,voltage , flux etc.) in dq (Park reference frame) needed to be transformed further through network (Kron reference frame) for multi-machine dynamic analysis ?

[3]

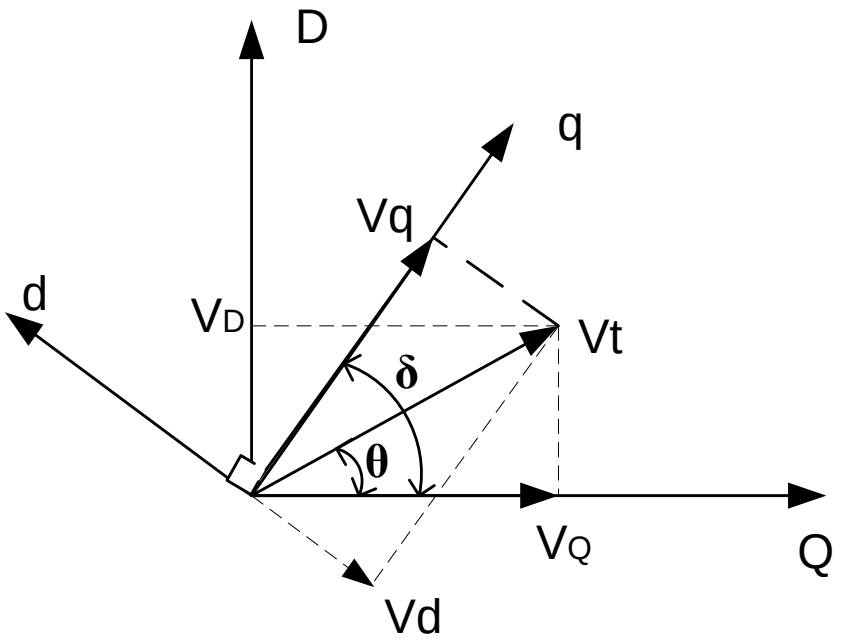
Ans: The Park transformation is restricted to a single generator with its terminal voltage as reference. In a multi-machine system , those machine terminal voltage angles are expressed with respect to one particular bus voltage angle . So each terminal voltage angle can not be treated as reference . The quantities obtained through Park Transformation in relation to an individual machines need to be adjusted for this voltage angle with respect to system voltage or network voltage angle reference . That reference is normally the slack bus angle (set to zero) used in power flow studies.

ii) Establish the relationship between the voltage between the two reference frames for machine load angle δ and machine terminal voltage (V_t) angle θ when they are with respect to the network reference frame.

[5]

Ans : In machine modeling we have laid the direct axis along the field. In practical power system there is large number of machines. Obviously one needs to have a common reference frame and orientation for synchronously rotating reference and express all the machines with reference to that reference frame. This is usually done by defining a network reference frame. Mathematically it is a transformation which can be constructed out of basic knowledge of trigonometry.

Let us assume that the network reference frame is expressed as DQ and that of machine as dq. If V is a voltage of a bus with an angle θ with respect to DQ reference; its corresponding component in machine reference frame would be a rotation of (δ-θ).



Synchronous reference frame with respect to network reference fram

$$VQ + j\,VD = Vt\,e^{j\theta}$$
$$VQ = Vt\cos\theta$$
$$VD = Vt\sin\theta$$

Resolving VQ and VD along machine reference d-q will produce

$$Vq = VQ\cos\delta + VD\sin\delta$$
$$Vd = -VQ\sin\delta + VD\cos\delta$$
$$Vq + j\,Vd = Vt\,e^{-j(\delta-\theta)}$$

iii) For $\delta = 30^{0}$ and $\theta = 5^{0}$ find out the d and q components of line terminal voltage of 20 kV in machine reference farme.

[3]

Ans : Vt = 20, $\delta = 30\ degrees$ and $\theta = 5\ degrees$ vq= 20*cos(25)= 18.12 kV, vd= 20*cos(25)= -8.45 kV.

iv) What are the purposes amortissuer windings in synchronous machine and how are they modelled for stability studies?

[2+2]

Ans: When the rotor swings , it deviates from synchronous speed . The swings are undesirable , so they are needed to be controlled/contained. The swing also is manifestation of extra energy . The rotor being a solid body, voltage induced and currents flow inside of it . That produces torque to counter or dampen the swing . At times , that measure from the solid rotor is not adequate , so additional closed circuit (copper rings) are placed on the rotor body, surface as well. They provide additional paths for additional currents that produce additional counter torque to dampen swing more effectively . One European

engineer D Ammortissuer invented this measure , so these windings are known as `ammortissuer windings` . They are modelled as differential equations with no input /impressed voltage as they are short-circuited winding in electrical as well as in mechanical sense .

b)

i) Why is neither caged rotor (Type 1) nor wound rotor (Type 2) wind turbine generator (WTG) the present preferred choice of the industry?

[4]

Ans: Neither of them offers greater wind speed variation . The efficiency of power production is low. For large rated machine , there are more copper losses at higher sleep , which heats up the machine. They need continuous source of magnetic flux , so capacitors are needed . The maintenance of capacitor bank is issue . Type 2 also needs brush sleep ring in the rotor . Rotor current control is not possible in Type 1 at all . Type 2 has some room for variation in rotor resistance – none of them are applicable for MPPT as slip is positive at less than synchronous speed.

ii) What is stall in respect of WTG operation ?

[3]

Ans: Stall is the consequence of reduced lift force on the blade because of the change of the angle of attack of the wind to the blade. The velocity distribution of moving air over the top blade surface also have some turbulence also, not helping the lift force because there is less and shorter duration contact with blade surface . When the angle of attack is changed to face more wind , lift force drops , the blade started moving at lower speed because of lower torque. This is stall. When it can be produced by changing angle of attack, it is known as active stall. Because of high wind speed the velocity distribution of wind on two sides of the blades starts becoming turbulent , the net lift is reduced . It is the curvature and geometry of the blade that produces the effect without physical movement of the blade around its axis . This is passive stall.

iii) `Some types of stall controls are in practice and some types are discontinued` – Justify this with reasons.

[3]

Ans: Active stall requires controller and actuator arrangements . It also causes stress on the bearing support system and the blade hub. It is not beneficial for operating life of the WT structure. So it is discontinued . Passive stall is continued because it is the geometry of the blade that takes care of it without any stress on the structure.

[25 marks]

4.

a)

i) What is the capacity factor of a wind turbine?

[2]

Ans : It is the ratio between actual energy produced and maximum possible energy that could have been produced based on rated capacity over a year.

ii) Why do offshore wind turbines have larger capacity factor over their onshore counterpart?

[2]

Ans: The wind speed in offshore is more consistent than that of offshore , it is most laminar than that on onshore.

iii) What is the tip-speed ratio of a wind turbine generator ?

[3]

Ans: The ratio between the linear velocity of the blade tip to the velocity of the wind .

iv) How does the tip-speed ratio help in optimal power control under varying wind speed ?

[3]

Ans : When tip speed ratio (TSR) is varied , the power conversion cofficent varies , at a particular value of TSR conversion co-efficient is maximum – this is known as optimal TSR. Normally it is between 6-8.

v) Why is the turbine side mechanical control adopted only above the rated wind speed ?

[2]

Ans: The mechanical side control is set in such a way that they are not exercised much upto the rated wind speed – this avoids stress on the hub and bearings . Above the rated wind speed this option is exercised not produce more than rated to protect the electrical side from over current and also mechanical structure from the stress.

b)

i) What is the advantage of Type 4 WTGs over their Type 3 counter parts from the dynamics and control point of view

[4]

Ans: Type 4 does not involve power collection from rotor. Gear box can be optional, low frequency voltage at the stator is advantage from efficiency point of view . There is complete disconnection between network and rotor side because of the conveter.

ii) How are pitch control WTGs modelled for stability studies ?

[2]

Ans: Usually by first order transfer function with pitch rate limiter with max and min pitch angle limit; There should also be a dead-zone included at the output of the controller in order not to let the pitch control mechanism to be active on sudden and temporary wind speed change. The pitch should only be allowed to work when change in wind speed is permanent to avoid any unnecessary wear tear of the blade housing and bearings .

iii) `Within a wind farm AC network, the oscillatory stability is more important than transient stability' – Justify this statement with reasons.

[3]

Ans: The transient stability in rotor angle sense is less of a concern in the wind farm power network . This is because the network is formed by conveter supported voltage /current sources . The acceleration of rotor mass is managed well through crow bar type protection where over voltage in DC link capacitors can be controlled through chopper/bleeding resistance, stator current can be controlled to control electromagnetic torque to move from optimal TSR. So loss of synchronisation of the rotor with respect to the network or power system is not an issue – which is the transient stability in classical sense. The wind farm network will have inductance, capacitance where there will be more oscilaltions which will be linked to control of the converters and interactions amongst them through network. Individual converter can dropp of the network , that is synchronisation issue in stationary sense where no physical movement is involved. So it is not a transient stability problem. But the elctromagnetic transients (EMT) are very important.

iv) How is *wake* in wind farm modelled for stability studies ?

[2]

Ans: For power system stability studies , the wake is modelled as a fractional drop in wind speed. There are several (Jensen etc ,) nonlinear algebraic models. For blade's structural stability studies , finite element models are appropriate.

v) How does *wake* impact the frequency stability capability and delivery of a wind farm?

[2]

Ans : The wake always affects the power output negatively , so frequency stability capability is impacted accordingly. There will always be mismatch between committed power ramping to actual delivery. This is not possible to have a precise calculation of the total volume as every single WTG in the wind farm will be subjected to wake that is unpredictably non uniform. Overall wake impacts negatively on this capability.

[25 marks]